

Mesh Enterprise Platform — Enterprise Documentation Suite

Version 1.0 — Enterprise Certification Sprint Classification: Customer-Facing

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1. Deployment Guide

1.1 System Requirements

Component	Minimum	Recommended
CPU	4 vCPU	8 vCPU
RAM	8 GB	16 GB
Disk	50 GB SSD	100 GB SSD
OS	Ubuntu 22.04 LTS	Ubuntu 24.04 LTS
Node.js	20.x LTS	22.x LTS
MySQL	8.0	8.0+ or TiDB
Network	100 Mbps	1 Gbps

1.2 Docker Compose Deployment (Recommended)

This is the fastest path to a running instance. Requires Docker 24+ and Docker Compose v2.

```
# 1. Clone the repository
git clone https://github.com/your-org/agentthinkmesh-full.git
cd agentthinkmesh-full

# 2. Configure environment
cp .env.template .env
vi .env # Fill in all CHANGE_ME values

# 3. Start all services
docker compose up -d

# 4. Verify health
curl http://localhost:3000/api/health
# Expected: {"status":"ok","db":"connected","version":"1.0.0","uptime":...
```

The `docker-compose.yml` starts three services:

- **app** — the Node.js application (port 3000)
- **db** — MySQL 8.0 (port 3306, internal only)
- **adminer** — Database admin UI (port 8080, disable in production)

1.3 Bare-Metal / VM Deployment

For environments where Docker is not available:

```
# Prerequisites
node --version    # Must be 20+
npm install -g pnpm@9 pm2

# Deploy
cp .env.template .env && vi .env
chmod +x scripts/start-onprem.sh
./scripts/start-onprem.sh

# Verify
curl http://localhost:3000/api/health
pm2 status
```

The startup script performs: env validation → dependency install → DB migration → build → PM2 start → health check.

1.4 Environment Variables

All required variables are documented in `.env.template`. Critical variables:

Variable	Description	Required
DATABASE_URL	MySQL connection string	Yes
JWT_SECRET	32+ character random secret for session signing	Yes
ANTHROPIC_API_KEY	Claude API key for council engine	Yes
BUILT_IN_FORGE_API_KEY	Manus platform API key	Yes
VITE_APP_ID	OAuth application ID	Yes
ENCRYPTION_MASTER_KEY	32-byte hex key for CMK encryption	Yes
PORT	HTTP port (default: 3000)	No

1.5 SSL/TLS Configuration

The platform does not terminate TLS directly. Place a reverse proxy (nginx, Caddy, AWS ALB) in front:

nginx example:

```
server {  
    listen 443 ssl;  
    server_name mesh.yourcompany.com;  
    ssl_certificate      /etc/ssl/mesh.crt;  
    ssl_certificate_key  /etc/ssl/mesh.key;  
    ssl_protocols        TLSv1.2 TLSv1.3;  
    ssl_ciphers           HIGH:!aNULL:!MD5;  
  
    location / {  
        proxy_pass http://localhost:3000;  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
    }  
}
```

1.6 Upgrade Procedure

```
# 1. Pull latest code  
git pull origin main  
  
# 2. Zero-downtime reload (PM2)  
./scripts/start-onprem.sh --reload  
  
# OR full restart (Docker)  
docker compose pull && docker compose up -d
```

1.7 Rollback Procedure

```
# Identify the last stable commit
git log --oneline -10

# Rollback to that commit
git checkout <commit-hash>
./scripts/start-onprem.sh --reload

# Rollback database (if schema changed)
# Restore from backup taken before upgrade (see Section 5)
```

2. Administration Guide

2.1 First-Time Setup

After deployment, the platform owner (the account whose `OWNER_OPEN_ID` is set in `.env`) has full administrative access.

Initial steps:

1. Log in at `https://your-domain/` using Manus OAuth
2. Navigate to **Enterprise** → **Organizations**
3. Create your first organisation
4. Add users via **Enterprise** → **User Management**
5. Assign roles
6. Create Decision Twins via **Admin** → **Twin Generator**

2.2 Organisation Management

Organisations are the top-level tenant unit. Each organisation is fully isolated from all others.

Create an organisation:

- Navigate to **Enterprise** → **Organizations** → **New Organisation**
- Enter name, slug (URL-safe identifier), and plan tier
- The slug cannot be changed after creation

Organisation plans:

Plan	Max Users	Max Twins	Council Modes
starter	5	3	Standard
professional	25	10	Standard + GCC
enterprise	Unlimited	Unlimited	All modes

2.3 User Management

Add a user:

1. Navigate to **Enterprise** → **User Management** → **Invite User**
2. Enter the user's Manus account email
3. Select their role
4. Click **Send Invitation**

Roles:

Role ID	Name	Capabilities
1	org_admin	Full org management, user management, twin management
2	dept_head	Department management, twin runs, reports
3	analyst	Twin runs, view reports, outcome recording
4	viewer	Read-only access to reports and outcomes

Suspend a user:

- Navigate to **Enterprise** → **User Management**
- Find the user → **Actions** → **Suspend**
- The user is immediately locked out; their data is preserved

Reactivate a user:

- Find the suspended user → **Actions** → **Reactivate**

2.4 Decision Twin Management

Create a Decision Twin:

- 1. Navigate to **Admin** → **Twin Generator**
- 2. Select a blueprint (M&A Screener, Capital Allocator, Vendor Risk, etc.)
- 3. Configure: display name, governance profile, industry, geography
- 4. Click **Generate Twin**
- 5. The twin appears in the Enterprise Dashboard

Governance profiles:

Profile	Description
STANDARD	Default — balanced disclosure
CONFIDENTIAL	Restricted — session data not shared across departments
RESTRICTED	Maximum — admin-only access to session history

Configure connectors:

- 1. Navigate to **Admin** → **Registry** → select a twin
- 2. Click **Connectors** → **Add Connector**
- 3. Select type: CSV, Excel, REST API, SQL
- 4. Configure connection parameters
- 5. Click **Test Connection** → **Save**

2.5 Platform Administration

Health monitoring:

```
# Check platform health
curl https://your-domain/api/health

# PM2 process monitor
pm2 monit

# View application logs
pm2 logs mesh-enterprise --lines 100
```

Database administration:

- Access Adminer at `http://localhost:8080` (Docker) or use any MySQL client
- Connection details are in `.env` under `DATABASE_URL`

3. Operations Guide

3.1 Daily Operations Checklist

Task	Frequency	Method
Health check	Every 5 min (automated)	<code>/api/health</code> endpoint
Log review	Daily	<code>pm2 logs</code>
Backup verification	Daily	See Section 5
Disk space check	Weekly	<code>df -h</code>
Certificate expiry	Monthly	<code>openssl x509 -enddate</code>
Dependency audit	Monthly	<code>pnpm audit</code>

3.2 Monitoring

The `/api/health` endpoint returns:

```
{
  "status": "ok",
  "db": "connected",
  "version": "1.0.0",
  "uptime": 86400,
  "timestamp": "2026-08-02T07:00:00.000Z"
}
```

Integrate with your monitoring tool (Datadog, Prometheus, UptimeRobot) by polling this endpoint every 60 seconds. Alert if status is not `"ok"` or if HTTP response code is not 200.

3.3 Log Management

Logs are written to `./logs/` by PM2:

- `logs/app-out.log` — application stdout
- `logs/app-err.log` — application stderr
- `logs/app-combined.log` — combined log

Log rotation is configured in `ecosystem.config.cjs` (10 MB max, 30-day retention).

For centralised logging, configure a log shipper (Filebeat, Fluentd) to forward `logs/*.log` to your SIEM or log aggregation platform.

3.4 Performance Tuning

Node.js heap size (for high-volume deployments):

```
# In ecosystem.config.cjs, add to node_args:
node_args: "--max-old-space-size=4096"
```

Database connection pool: The platform uses a single connection per request (Drizzle ORM). For >50 concurrent users, configure a connection pooler (PgBouncer equivalent for MySQL: ProxySQL).

Council execution timeout: Council runs take 60–120 seconds. The default Express request timeout is 180 seconds. Do not reduce below 180 seconds.

4. Security Guide

4.1 Authentication

The platform uses Manus OAuth 2.0 for authentication. All sessions are signed JWT cookies with:

- Algorithm: HS256
- Expiry: 7 days
- HttpOnly: true
- SameSite: Lax
- Secure: true (production)

Air-gapped environments: OAuth requires outbound connectivity to `api.manus.im`. For fully air-gapped deployments, contact Manus for an on-premises OAuth server package.

4.2 Authorisation

All tRPC procedures are protected at three levels:

1. **Authentication gate** — `protectedProcedure` rejects unauthenticated requests with HTTP 401
2. **Tenant isolation** — `requireOrgMembership` middleware resolves `orgId` from the authenticated user's actual membership record; it never trusts client-supplied `orgId` values
3. **Role gate** — individual procedures check `ctx.user.role` or `ctx.membership.roleId` for privileged operations

Cross-tenant access is impossible by design. The `requireOrgMembership` middleware throws `FORBIDDEN` if the user is not an active member of the organisation.

4.3 Tenant Isolation

Each organisation's data is isolated at the application layer:

- All database queries include `WHERE orgId = ctx.orgId`
- `ctx.orgId` is resolved exclusively from the authenticated user's membership record
- No client-supplied `orgId` is trusted for data access
- Cross-tenant penetration tests (20 scenarios) pass in the CI suite

4.4 Encryption

Data at rest:

- Customer Master Key (CMK) system encrypts sensitive fields using AES-256
- Keys are stored in the `encryptionKeys` table, encrypted with the `ENCRYPTION_MASTER_KEY` env var
- Key rotation is supported without downtime

Data in transit:

- All traffic must be served over TLS 1.2+ (enforced by HSTS header)
- The platform sets `Strict-Transport-Security: max-age=31536000; includeSubDomains; preload`

4.5 Security Headers

The following headers are set on every response (via Helmet):

Header	Value
<code>Strict-Transport-Security</code>	<code>max-age=31536000; includeSubDomains; preload</code>
<code>X-Content-Type-Options</code>	<code>nosniff</code>
<code>X-Frame-Options</code>	<code>DENY</code>
<code>Referrer-Policy</code>	<code>strict-origin-when-cross-origin</code>
<code>Content-Security-Policy</code>	Restrictive policy (see source)
<code>Permissions-Policy</code>	<code>camera=(), microphone=(), geolocation=()</code>
<code>X-DNS-Prefetch-Control</code>	<code>off</code>
<code>X-Download-Options</code>	<code>noopen</code>

4.6 Rate Limiting

Endpoint Group	Limit	Window
All <code>/api/*</code> routes	500 requests	15 minutes per IP
<code>/api/oauth/*</code> routes	20 requests	15 minutes per IP
<code>/api/health</code>	Unlimited	—

4.7 Audit Logging

All enterprise actions are written to the `auditLogs` table:

- Twin runs and simulations
- Member suspension and reactivation
- Role changes
- Connector configuration changes
- Report generation

Audit logs are append-only and include: timestamp, userId, orgId, action, resourceType, resourceId, metadata.

4.8 Secrets Management

- All secrets are injected via environment variables
- No secrets are committed to the repository
- Use `.env.template` as the reference; never commit `.env`
- Rotate `JWT_SECRET` by updating the env var and restarting the server (existing sessions will be invalidated)
- Rotate `ENCRYPTION_MASTER_KEY` using the key rotation runbook (contact support)

4.9 On-Premises Security Hardening

For on-premises deployments:

1. Run the application as a non-root user (`useradd -m mesh && su mesh`)
 2. Restrict database access to localhost only
 3. Enable firewall: allow only ports 443 (HTTPS) and 22 (SSH admin)
-

- 4. Disable Adminer in production (`docker compose --profile prod up -d`)
- 5. Enable OS-level audit logging (auditd)
- 6. Configure log forwarding to your SIEM

5. Backup and Recovery Guide

5.1 Backup Strategy

Component	Method	Frequency	Retention
Database	<code>mysqldump</code>	Daily (automated)	30 days
Application config	<code>.env</code> file backup	On every change	10 versions
Uploaded files	S3 cross-region replication	Continuous	90 days
Application code	Git repository	Every commit	Indefinite

5.2 Database Backup

Manual backup:

```
# Extract connection details from DATABASE_URL
# Format: mysql://user:password@host:port/database

mysqldump \
  --host=<host> \
  --port=<port> \
  --user=<user> \
  --password=<password> \
  --single-transaction \
  --routines \
  --triggers \
  <database> \
  > backup-$(date +%Y%m%d-%H%M%S).sql

gzip backup-*.sql
```

Automated backup (cron):

```
# Add to crontab: crontab -e
0 2 * * * /home/mesh/agentthinkmesh-full/scripts/backup-db.sh >> /var/log/
```

Create `scripts/backup-db.sh` :

```
#!/usr/bin/env bash
set -euo pipefail
source /home/mesh/agentthinkmesh-full/.env
BACKUP_DIR="/var/backups/mesh"
mkdir -p "$BACKUP_DIR"
FILENAME="mesh-db-$(date +%Y%m%d-%H%M%S).sql.gz"

# Parse DATABASE_URL: mysql://user:pass@host:port/db
DB_USER=$(echo "$DATABASE_URL" | sed 's|mysql://\[^\:]*\):.*|\1|')
DB_PASS=$(echo "$DATABASE_URL" | sed 's|mysql://\[^\:]*\:\[^\@]*\)|@.*|\1|')
DB_HOST=$(echo "$DATABASE_URL" | sed 's|.*@\([^\:]*\):.*|\1|')
DB_PORT=$(echo "$DATABASE_URL" | sed 's|.*:\([0-9]*\)\/.*|\1|')
DB_NAME=$(echo "$DATABASE_URL" | sed 's|.*\/\([^\?]*\)\/.*|\1|')

mysqldump --host="$DB_HOST" --port="$DB_PORT" --user="$DB_USER" \
  --password="$DB_PASS" --single-transaction --routines --triggers \
  "$DB_NAME" | gzip > "$BACKUP_DIR/$FILENAME"

# Delete backups older than 30 days
find "$BACKUP_DIR" -name "*.sql.gz" -mtime +30 -delete

echo "[$(date)] Backup complete: $FILENAME"
```

5.3 Recovery Procedure

Full database recovery:

```
# 1. Stop the application
./scripts/start-onprem.sh --stop

# 2. Restore the database
gunzip -c backup-20260802-020000.sql.gz | \
  mysql --host=<host> --user=<user> --password=<password> <database>

# 3. Restart the application
./scripts/start-onprem.sh

# 4. Verify health
curl http://localhost:3000/api/health
```

Point-in-time recovery (MySQL binary logs): If binary logging is enabled on your MySQL server, you can recover to any point in time. Contact your DBA for the `mysqlbinlog` procedure.

5.4 Disaster Recovery

RTO (Recovery Time Objective): 4 hours

RPO (Recovery Point Objective): 24 hours (daily backup) / 1 hour (with binary logs)

DR runbook:

1. Provision a new server matching the system requirements
 2. Clone the repository and restore `.env`
 3. Restore the latest database backup
 4. Run `./scripts/start-onprem.sh`
 5. Update DNS to point to the new server
 6. Verify all functionality via the health endpoint and a test council run
-

6. Troubleshooting Guide

6.1 Common Issues

Platform does not start

Symptom: `./scripts/start-onprem.sh` exits with an error.

Diagnosis:

```
# Check Node.js version
node --version    # Must be 20+

# Check .env file
cat .env | grep -v "^#" | grep -v "^$"

# Check database connectivity
mysql -h <host> -u <user> -p<password> -e "SELECT 1"

# Check PM2 logs
pm2 logs mesh-enterprise --lines 50
```

Common causes:

- Missing or incorrect `DATABASE_URL` — verify the connection string format
- `JWT_SECRET` too short — must be 32+ characters
- Node.js version < 20 — upgrade Node.js
- Port 3000 already in use — set `PORT=3001` in `.env`

Health check returns `"db": "error"`

Symptom: `GET /api/health` returns `{"status": "degraded", "db": "error"}`

Diagnosis:

```
# Test database connection directly
mysql -h <host> -u <user> -p<password> <database> -e "SELECT COUNT(*) FROM"
```

Common causes:

- Database server is down or unreachable
- Firewall blocking port 3306
- Incorrect credentials in `DATABASE_URL`
- Database user lacks SELECT privileges

Council run times out

Symptom: Twin run returns an error after ~180 seconds.

Diagnosis:

- Check `ANTHROPIC_API_KEY` is valid: `curl -H "x-api-key: $ANTHROPIC_API_KEY" https://api.anthropic.com/v1/models`
- Check outbound internet connectivity: `curl -I https://api.anthropic.com`
- Check the application logs for the specific error: `pm2 logs mesh-enterprise | grep -i "council\\|anthropic\\|timeout"`

Common causes:

- Invalid or expired `ANTHROPIC_API_KEY`
- Outbound HTTPS blocked by firewall (requires `api.anthropic.com:443`)
- Anthropic API rate limit exceeded

Users cannot log in

Symptom: Login redirects to an error page or loops.

Diagnosis:

```
# Check OAuth configuration
echo "App ID: $VITE_APP_ID"
echo "OAuth URL: $OAUTH_SERVER_URL"

# Check server logs for OAuth errors
pm2 logs mesh-enterprise | grep -i "oauth\\|callback\\|token"
```

Common causes:

- `VITE_APP_ID` does not match the registered OAuth application
- The callback URL registered in the OAuth portal does not match the deployed domain
- Clock skew > 5 minutes between server and OAuth provider (sync NTP)

“FORBIDDEN” error on enterprise procedures

Symptom: Enterprise API calls return `TRPC_ERROR: FORBIDDEN`.

Cause: The user is not an active member of the organisation. This is the tenant isolation middleware working correctly.

Resolution:

- Verify the user has been added to the organisation via **Enterprise** → **User Management**
- Verify their membership status is `active` (not `suspended`)
- If the user was recently added, ask them to log out and log back in

6.2 Log Reference

Log Pattern	Meaning
<code>[OAuth] Initialized</code>	Server started successfully
<code>[Health] DB connected</code>	Database connection healthy
<code>[Council] Starting run</code>	Council execution beginning
<code>[Council] Completed in Xms</code>	Council execution successful
<code>[Tenant] FORBIDDEN</code>	Cross-tenant access attempt blocked
<code>[Rate Limit] Too many requests</code>	Rate limit triggered

6.3 Support Escalation

For issues not resolved by this guide:

1. Collect logs: `pm2 logs mesh-enterprise --lines 200 > support-bundle.txt`
2. Include the output of `curl http://localhost:3000/api/health`
3. Include the `.env` file with all secret values redacted
4. Submit to support with the subject: `[Enterprise Support] <issue description>`

7. Pilot Onboarding Guide

7.1 Pre-Pilot Checklist

Before the pilot begins, verify:

- ☐ Platform deployed and health check passing
- ☐ SSL certificate installed and HTTPS working
- ☐ At least one organisation created
- ☐ Admin user added and can log in
- ☐ At least one Decision Twin created and visible in dashboard
- ☐ Council run test completed successfully
- ☐ Backup procedure tested and verified

7.2 Day 1: Administrator Onboarding (2 hours)

Hour 1: Platform walkthrough

1. Log in and review the dashboard
2. Navigate to **Enterprise** → **Organizations** — review org settings
3. Navigate to **Enterprise** → **User Management** — add pilot users
4. Navigate to **Admin** → **Twin Generator** — create the first pilot twin

Hour 2: First council run

1. Open the Enterprise Dashboard
2. Select the Decision Twin
3. Click **Run** — enter a real business question
4. Review the council output: verdict, score, conditions, dissents
5. Review the outcome in **Outcome Ledger**

7.3 Week 1: Pilot Scope (Alghanim Industries Reference)

Recommended pilot configuration:

Element	Specification
Organisation	Alghanim Industries — Pilot
Departments	Corporate Development, Treasury
Users	2 admins, 4 analysts, 2 viewers
Decision Twins	M&A Screener, Capital Allocator, Vendor Risk
Council mode	GCC
Governance profile	CONFIDENTIAL
Session target	10 real decisions in Week 1

Week 1 decision questions (examples):

- “Should we acquire a 35% stake in [company] for [amount]?”
- “Should we allocate [amount] to [asset class] given current market conditions?”
- “Should we renew the [vendor] contract at the proposed terms?”

7.4 Success Metrics

Metric	Target	Measurement
Council run completion rate	>95%	Sessions completed / sessions started
Decision quality score	>0.70 average	<code>finalScore</code> in outcome ledger
Time to verdict	<3 minutes	Session duration
User adoption	>80% weekly active	Users with ≥ 1 session per week
System availability	>99.5%	Health check uptime
User satisfaction	> $\frac{4}{5}$	End-of-pilot survey

7.5 Pilot Exit Criteria

The pilot is considered successful when:

1. At least 20 council sessions completed
 2. Average decision quality score > 0.70
 3. At least 3 decisions acted upon by management
 4. Zero security incidents
 5. System availability > 99.5%
 6. User satisfaction score > $\frac{4}{5}$
-

8. Deployment Checklist

Pre-Deployment

- ☐ Server provisioned and meets minimum requirements
- ☐ Node.js 20+ installed
- ☐ MySQL 8.0 database provisioned and accessible
- ☐ All environment variables in `.env` filled in (no `CHANGE_ME` values remaining)
- ☐ `node scripts/validate-env.js` passes without errors
- ☐ SSL certificate obtained and configured in reverse proxy
- ☐ DNS record pointing to the server
- ☐ Firewall configured: 443 inbound, 3306 localhost-only
- ☐ Backup storage configured and accessible

Deployment

- ☐ `./scripts/start-onprem.sh` completes without errors
- ☐ `GET /api/health` returns `{"status":"ok","db":"connected"}`
- ☐ PM2 shows `mesh-enterprise` as `online`
- ☐ Application accessible at `https://your-domain/`
- ☐ Login flow completes successfully
- ☐ First organisation created
- ☐ First admin user added
- ☐ First Decision Twin created

- ☐ Test council run completes (< 3 minutes)
- ☐ Test outcome recorded in Outcome Ledger

Post-Deployment

- ☐ Monitoring configured (health endpoint polled every 60s)
- ☐ Alerting configured (notify on health check failure)
- ☐ First database backup taken and verified
- ☐ Backup restoration tested on a separate server
- ☐ Log forwarding configured (if required)
- ☐ PM2 startup on boot configured: `pm2 startup && pm2 save`
- ☐ Security headers verified: `curl -I https://your-domain/ | grep -i "strict\|content-security\|x-frame"`
- ☐ Rate limiting verified: 20 rapid requests to `/api/oauth/` triggers 429
- ☐ Pilot users onboarded and trained
- ☐ Support contact established

Go-Live Sign-Off

Role	Name	Date	Signature
IT Operations Lead			
CISO / Security Lead			
Business Sponsor			
Platform Administrator			

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